

- 1 (a) The Earth may be considered to be a uniform sphere of radius $6.38 \times 10^6 \text{ m}$. Its mass is assumed to be concentrated at its centre.

Given that the gravitational field strength at the Earth's surface is 9.81 N kg^{-1} , show that the mass of the Earth is $5.99 \times 10^{24} \text{ kg}$.

[2]

- (b) A satellite is placed in geostationary orbit around the Earth.

- (i) Calculate the angular speed of the satellite in its orbit.

angular speed = rad s^{-1} [3]

- (ii) Using the data in (a), determine the radius of the orbit.

radius = m [3]